7th & 8th grade

Courses

Subject Area	Course Name	Term Period
Algebra I	Algebra I	Full Year
Geometry	Geometry	Full Year
Yr 4+ Lang Other Than English	Chinese	Full Year

Has International Experience **Yes, English**

High schools

YK PAO SCHOOL
CHINA 201620, CHINA, PEOPLES REPUBLIC OF

School code	**694616**
Specialized curriculum	**International Baccalaureate**
Grading system	**1-7 Int'l Bacc** **IGCSE, P/NP**
Term system	**Full**
Attended	**August 2019 - May 2023**
Grades attended	**9th grade** 2019 - 2020 **10th grade** 2020 - 2021 **11th grade** 2021 - 2022 **12th grade** 2022 - 2023
School you will graduate from:	**Yes**
Degree, diploma or certificate	**IB Diploma**
Date received or to be received	**May 2023**
Language of Instruction	**English**

9th grade

YK PAO SCHOOL
CHINA 201620, CHINA, PEOPLES REPUBLIC OF

2019 - 2020 academic year

Subject area/Course category	Course name	Hnrs	Term	GRDG System	G1	G2	G3	G4	Language name
A-World History/ Cultures/ Hist. Geography	9 Geography	NH	Full	Other	A*				
B-English	9 English	NH	Full	Other	A				
C-Mathematics II	9 Mathematics	NH	Full	Other	A*				
D-Chemistry	9 Chemistry	NH	Full	Other	A*				
D-Physics	9 Physics	NH	Full	Other	A*				
E-LOTE Level 4+	9 Chinese	NH	Full	Other	A				Chinese
G-History / Social Science	9 Economics	NH	Full	Other	A*				

10th grade

YK PAO SCHOOL
CHINA 201620, CHINA, PEOPLES REPUBLIC OF

2020 - 2021 academic year

Subject area/Course category	Course name	Hnrs	Term	GRDG System	G1	G2	G3	G4	Language name
A-World History/ Cultures/ Hist. Geography	IGCSE Geography	NH	Full	Other	A*				
B-English	IGCSE First Language English	NH	Full	Other	B				
B-English	IGCSE Literature in English	NH	Full	Other	B				
C-Mathematics III	IGCSE Additional Maths	NH	Full	Other	A*				
D-Chemistry	IGCSE Chemistry	NH	Full	Other	A*				
D-Physics	IGCSE Physics	NH	Full	Other	A*				
E-LOTE Level 4+	IGCSE Chinese	NH	Full	Other	A				Chinese
G-History / Social Science	IGCSE Economics	NH	Full	Other	A*				

11th grade

YK PAO SCHOOL
CHINA 201620, CHINA, PEOPLES REPUBLIC OF

2021 - 2022 academic year

Subject area/Course category	Course name	Hnrs	Term	GRDG System	G1	G2	G3	G4	Language name
B-English	IB English A SL	IB	Full	1-7 Int'l Bacc	6				
C-Other Advanced Mathematics	IB Mathematics AA HL	IB	Full	1-7 Int'l Bacc	7				
D-Computer Science	IB Computer Science HL	IB	Full	1-7 Int'l Bacc	7				
D-Physics	IB Physics HL	IB	Full	1-7 Int'l Bacc	7				
E-LOTE Level 4+	IB Chinese A SL	IB	Full	1-7 Int'l Bacc	7				Chinese
G-History / Social Science	IB Economics SL	IB	Full	1-7 Int'l Bacc	7				
G-Interdisciplinary	Theory of Knowledge	IB	Full	Other	Pass				

12th grade

YK PAO SCHOOL
CHINA 201620, CHINA, PEOPLES REPUBLIC OF

2022 - 2023 academic year

Subject area/Course category	Course name	Hnrs	Term	GRDG System	G1	G2	G3	G4	Language name
B-English	IB English A SL	IB	Full	1-7 Int'l Bacc	IP				
C-Other Advanced Mathematics	IB Mathematics AA HL	IB	Full	1-7 Int'l Bacc	IP				
D-Computer Science	IB Computer Science HL	IB	Full	1-7 Int'l Bacc	IP				
D-Physics	IB Physics HL	IB	Full	1-7 Int'l Bacc	IP				
E-LOTE Level 4+	IB Chinese A SL	IB	Full	1-7 Int'l Bacc	IP				Chinese
G-History / Social Science	IB Economics SL	IB	Full	1-7 Int'l Bacc	IP				
G-Interdisciplinary	Theory of Knowledge	IB	Full	Other	IP				

Colleges attended while in high school

OBERLIN COLG AND CONSERVATORY
OBERLIN, OH

College code:	001587
Grading system:	A B C D F
Attended:	June 2021 - September 2021
Grades attended:	11th grade: Summer 2021

College courses taken in high school

OBERLIN COLG AND CONSERVATORY
OBERLIN, OH

College code: 001587
Grading system: A B C D F

11th Grade - Summer 2021

Dept.	Course No.	Course title	Units	Honors status	Grade	Subject area	Language name
INST	099	Pioneer Research Course	2.0	CL	A	G-Interdisciplinary	

Additional information

Test Scores

SAT Subject Tests

I have no test score information to report

AP exams

I have no test score information to report

IB exams

Reported IB exams

Subject area	Level	Exam name	Date taken	Score	Planned date
Experimental Sciences	HL	PHYSICS			05/2023
Individuals and Society	SL	ECONOMICS			05/2023
Languages A1	SL	ENGLISH A LAL			05/2023

Subject area	Level	Exam name	Date taken	Score	Planned date
Languages A1	SL	CHINESE A LAL			05/2023
Mathematical and Computer Science	HL	COMPUTER SC.			05/2023
Mathematical and Computer Science	HL	MATHEMATICS ANALYSIS			05/2023

Reported predicted IB scores

Subject area	Level	Exam name	Planned date	Predicted score
Experimental Sciences	HL	PHYSICS	05/2023	7
Individuals and Society	SL	ECONOMICS	05/2023	7
Languages A1	SL	CHINESE A LAL	05/2023	7
Languages A1	SL	ENGLISH A LAL	05/2023	7
Mathematical and Computer Science	HL	COMPUTER SC.	05/2023	7
Mathematical and Computer Science	HL	MATHEMATICS ANALYSIS	05/2023	7

English language proficiency test

Reported test

Date taken November 2021
Test type TOEFL iBT
Registration number 2655511214722787

Subject	Score
Reading	30
Listening	29
Writing	26
Speaking	27
Total	112

International exams

GCSE/IGCSE/O-levels (Grade scale: A*-G, U)

Date taken	Subject	Score	Predicted	Not received
06/2020	International Mathematics	A*		
06/2021	Additional Mathematics	A*		
06/2021	Chemistry	A*		
06/2021	Economics	A*		
06/2021	First Language Chinese	A*		
06/2021	First Language English	A*		
06/2021	Geography	A*		
06/2021	Literature in English	A*		
06/2021	Physics	A*		

Activities & Awards

Activities/Awards

1. Extracurricular activity

Activity name

President of Track and Field Club

Activity description

I lead trainings twice a week to promote health and inclusive sports culture on campus. I contacted off-campus coaching resources to guide training since my school doesn't have a coach. Track & field is about surpassing our limits and challenging our bodies, and everyone can engage in it. That is why I'm working to build a varsity team.

Grade participation

10th grade
11th grade
12th grade

Time commitment

4.0 hours per week
30.0 weeks per year

2. Educational preparation program

Program name

Pioneer Academics

Program description

I worked one-on-one with Dr. Gavrus from UC Berkeley on the application of Fourier Analysis and produced a research paper "Analysis of periodicity in COVID-19 daily new cases using Discrete Fourier Transform" as a reference for making related policies. I self-studied differential equations and linear algebra, and received A for my grade.

Grade participation

11th grade

Time commitment

20.0 hours per week
10.0 weeks per year

3. Award or honor

Name of the award or honor

Gold & Essay Writing Award, National Economics Challenge

Level of recognition

National

Type of award

Academic

Grade level when awarded

11th grade

Award requirements

Each year, close to 10,000 students from across the nation take the chance to be tested on key micro and macroeconomic principles, as well as their knowledge of the world economy. The Adam Smith (AS) division is for advanced level competitors.

What you did to earn award

Apart from school courses, I self-learned economics from Coursera and Youtube for 3 months. My teammates and I wrote an economics paper "China, Globalization, and Economic Disparity". We won regional team gold and national top team in essay writing in the AS division

4. Award or honor

Name of the award or honor

Founder Award (G9-10) & Principal Award (G9-11)

Level of recognition

School

Type of award

Academic

Grade level when awarded

9th grade
10th grade
11th grade

Award requirements

The Founder Award is given to students in the top 10% of academic performance, including IG courses and courses for Chinese middle schools. The Principal Award is given to the top 20%. The Founder Award for G11 is yet to be decided.

What you did to earn award

I maintained excellent academic performance.

5. Award or honor

Name of the award or honor

Third Place, Real World Design Challenge China

Level of recognition

National

Type of award

Academic

Grade level when awarded

10th grade

Award requirements

The Real World Design Challenge (RWDC) provides high schoolers the opportunity to work on real world engineering challenges in a team environment. It's run by a public-private partnership with the goal of sustainably increasing the STEM workforce.

What you did to earn award

We wrote an investigation report on designing an Unmanned Arial System for drone package delivery. I am the project scientist, responsible for mathematically modeling the aerodynamics of the wings and creating an obstacle avoidance system demo using Python.

6. Award or honor

Name of the award or honor

Track Champion (800m), SSSA 2020

Level of recognition

State

Type of award

Non-academic

Grade level when awarded

9th grade
10th grade

Award requirements

Shanghai Sports School Association is the biggest organization in east China that provides competitive sporting events to schools in and around Shanghai. The best of the best athletes from high schools participate in these competitions.

What you did to earn award

As leader of my school's track & field club and winger of the soccer club, I train 2 hours twice a week in these two sports. Since my school doesn't have a track & field coach, I contacted Shanghai Youth Sports School and used their resources. My result: Track & Field U15 800m 1st place -123 seconds; Track & Field U15 400m 2nd place: 53.24 seconds

7. Award or honor

Name of the award or honor

Soccer Champion D1, Basketball Champion D2, SSSA 2020

Level of recognition

State

Type of award

Non-academic

Grade level when awarded

10th grade

Award requirements

Shanghai Sports School Association is the biggest sports league in east China that provides competitive sporting events to schools in and around Shanghai. The best of the best athletes from high schools participate in these competitions.

What you did to earn award

Long-term persistent training, skilled teamwork, and reasonable on-the-spot tactics are the keys to our victory. I was the MVP of the semifinals.

8. Extracurricular activity

Activity name

Member of 3 Varsity Teams (Soccer, Volleyball & Basketball)

Activity description

Awarded Most Improved Player of Varsity Boys Soccer and Most Valuable Player in Varsity Boys Volleyball; represented my school about 50 times (G10-12) in soccer, about 30 (G10-12) in Volleyball and about 60 (G9-10) in basketball; competed in SSSA and United World Games

Grade participation

9th grade
10th grade
11th grade
12th grade

Time commitment

15.0 hours per week
32.0 weeks per year

9. Volunteer / Community service

Organization, group or program name

Minzu Middle School in Ganluo County of Sichuan Province

Organization, group or program description

The headmaster of the school was a classmate of my father in primary school. 95% of its students are of Yi ethnicity and 55% students don't enter high school. Ganluo County was rated as one of the poorest areas in Sichuan Province.

Description of volunteer experience

I taught soccer and basketball. I believed that the passion developed from participating in sports will support them in difficulties and help them live a better life. There's also a chance they become successful athletes and help build Ganluo in the future. Chen, one of my students, was admitted to the Sichuan U15 Soccer Team.

Grade participation

10th grade
11th grade

Time commitment

30.0 hours per week
3.0 weeks per year

10. Educational preparation program

Program name

Independent Research

Program description

I learned from my volunteer teaching experience that sports education might provide another way for underprivileged kids to change their lives. I wrote an 8-page paper discussing whether the "Underdog Theory" is true and the potential of poverty alleviation through sports achievement. I submitted my paper to the County Mayor.

Grade participation

11th grade

Time commitment

5.0 hours per week
10.0 weeks per year

11. Volunteer / Community service

Organization, group or program name

Quarantine Fitness

Organization, group or program description

Quarantine fitness is an online workout community encouraging people to stay active in training. I founded the community during the 2-month Shanghai lockdown to ease people's pressure and promote a healthy lifestyle. 51 community members.

Description of volunteer experience

I suggested individualized workout methods, schedule, and diet plan to group members with different basic fitness skills through one on one calls. Then I organized daily video calls to give feedback and established weekly rankings to motivate them. We continued after the quarantine was over and we have accumulated more than 700 exercise hours.

Grade participation

11th grade

Time commitment

4.0 hours per week
24.0 weeks per year

12. Work experience

Company or organization name

Shanghai Yunyou Inc.

Company or organization description

Founded in 2018, the Yunyou inc. is a start-up producing wearable medical devices, such as real-time electrocardiograms (ECG) to diagnose heart diseases and prevent Injuries. With 22 employees, the company is valued at $50 million.

Job title

Engineer

Job responsibilities

I learned medical knowledge about the cardiovascular system and ECG. I worked on signal processing and using AI LSTM to identify wave forms in an ECG. I was interested in the device because the medical treatment resources could be scarce in rural areas, so the product of this company can help reduce the risk of severe diseases and injuries.

Grade participation and time commitment

11th grade
During summer 32.0 hours per week

Currently working at this job

No
Start Date: 07/2022
End Date: 08/2022

13. Award or honor

Name of the award or honor

Silver, USA Computing Olympiad

Level of recognition

International

Type of award

Academic

Grade level when awarded

10th grade

Award requirements

The bronze division had 7858 total participants, of whom 6127 were pre-college students. All competitors who scored 800 or higher on this contest (1312 of them, from the USA) are automatically promoted to the silver division.

What you did to earn award

I self-studied computer science by reading college textbooks. I did 8 practice problem sets from previous competitions.

14. Award or honor

Name of the award or honor

ARML Global Top 100 Teams

Level of recognition

International

Type of award

Academic

Grade level when awarded

10th grade

Award requirements

American Regional Mathematics League (ARML) held local competitions in China at the same time as the American contest. Participants solve problem sets to compare their results with their international peers. The top 100 scoring teams were awarded.

What you did to earn award

We participated in multiple rounds of competitions, including a team round, an individual round and tiebreaker question, and three relay rounds.

15. Volunteer / Community service

Organization, group or program name

Colloquium Cognitionis Callidum Program (CCC)

Organization, group or program description

CCC is founded by a few high school students that hold three camps: Sunflower, History & Humanities, and Science. Each is held once every year for 6 days. The camps are held for children of migrant workers in Shanghai.

Description of volunteer experience

I have been a lecturer and volunteer in the camp and spoke on topics including "Fourier Analysis", "Evolution of Human Body", "Machine Learning", "Life, Evolution and Entropy". I was also a cameraman and a volunteer providing logistical support.

Grade participation

9th grade
10th grade
11th grade
12th grade

Time commitment

20.0 hours per week
4.0 weeks per year

16. Educational preparation program

Program name

UPenn & ShanghaiTech U Summer

Program description

The program was designed to teach students to solve real world issues with latest technologies. I learned machine learning and coding during the camp and my team finished a project, "Garbage Classification Using Image Recognition", that used machine learning to help citizens in Shanghai classify garbage to fulfill government requirements.

Grade participation

9th grade

Time commitment

80.0 hours per week
1.0 weeks per year

17. Work experience

Company or organization name

Ziwen Education in Thames Town Shanghai

Company or organization description

Ziwen Education is a chain institution teaching kids traditional Chinese calligraphy, painting, chess etc. My classmates and I started a branch in Thames Town, right by my school and home.

Job title

Co-founder and Chief Financial Officer

Job responsibilities

In 2020, my two classmates and I sensed the business opportunity and started the company with the help from one of our parents. My job included making financial decisions, supervising accounting and writing financial reports. We later hired a manager. I used my dividend ($15K) to build basketball courts in Ganluo Middle School and for a new bike.

Grade participation and time commitment

9th grade
During school year 2.0 hours per week
During summer 10.0 hours per week
10th grade
During school year 2.0 hours per week
During summer 10.0 hours per week
11th grade
During school year 2.0 hours per week
During summer 10.0 hours per week
12th grade
During school year 2.0 hours per week
During summer 10.0 hours per week

Currently working at this job

Yes
Start Date: 07/2020

Work earnings

How have you used or will you use your earnings?

I earned 100K CNY ($15K) on Ziwen Education. I used it to buy an electric bike and clothes, and donated the rest to the primary school I volunteered at to build basketball courts. I haven't decided how to spend summer internship earnings (3000 CNY).

Scholarships & Programs

Scholarships

Academic major or interest

- Social sciences discipline

Ethnicity, national origin or religion

- From China or Hong Kong

Family circumstances and relationships

- Student whose first language is not English

School or geographic affiliation

- Attended Secondary School in China
- Resident of Beijing, Shanghai, Hangzhou, Shenzhen or Guangzhou

Support programs

None selected.

Personal Insight

Introduction

Personal insight questions

Describe an example of your leadership experience in which you have positively influenced others, helped resolve disputes, or contributed to group efforts over time.

I have a strong passion for the sport of sprinting and have recorded the fastest laps in my school. Therefore, in 11th grade, I became the organizer and leader of our school's track and field team. Last year, my goal was to win the Shanghai Sports Meet in November. However, the COVID-19 lockdown severely interrupted my training plan. During this 3-month lockdown, citizens were not allowed to leave their homes and almost all athletic facilities were closed. Though I was initially disappointed to see all of my preparations and systematic training become fruitless, I knew there would be more opportunities in the future and wanted to find a new outlet to keep active during the lockdown.

I decided to tidy up my dejected mood and make some active adaptations, especially as the lockdown period continued to extend. So, I first reached out to my friends to check on their fitness conditions and found that they were having similar concerns about the lack of physical activity. We were eager to work out, yet at the same time, many of my peers lacked the proper structure or plan to begin exercising at home.

To meet these needs and keep me motivated, I created an organization: Quarantine Fitness. Using my influence and leadership in school varsity sports, I was able to recruit a large group of peers to join and we quickly expanded to over fifty people. During this time, I developed new responsibilities: I set up video call fitness sessions, planned daily workout regimens that anyone could do at home, and established a weekly fitness ranking to incentivize and supervise members to work out regularly. I also invited my coach and athlete friends to the group, who suggested accessible diet plans and demonstrated correct workout movements to our members.

As our organization grew it became an essential part of many students' daily schedules. Together, we were able to gain health benefits from exercising and also received a mental health boost by exercising within a supportive community that facilitated socializing in quarantine.

Every person has a creative side, and it can be expressed in many ways: problem solving, original and innovative thinking, and artistically, to name a few. Describe how you express your creative side.

Growing up, I often found myself surrounded by hypocrisy rooted in Chinese culture. Parents educate us to cater to the outside world, school pressures us to hide our controversial and political opinions, and society impels us to fake feelings such as enthusiasm to be more sociable. But I don't accept that. I firmly believe that keeping true to myself is the right way to live. I feel compelled to challenge hypocrisy and express what's deep inside me. At my core is a desire for honesty.

That's why I began to write out my thoughts, hoping to strike at what's real from within. But I don't just write anything, I write lyrics for hip-hop songs. I was born in Chengdu, the center of Chinese hip-hop, which blossomed as it reproduced the works of American rap, encouraging youth to put their life and struggles to music. Therefore, I am deeply influenced by hip-hop culture, especially as it represents where I come from.

As I learned more about the Chengdu hip-hop culture, I realized that these famous rap artists that I look up to are ordinary humans as well. I was stunned to discover that most of them never received formal music education in universities; instead, they were just sharing their authentic life through rap and figuring out music production along the way. Their stories gave me the courage to start rapping as well. I

authentic life through rap and figuring out music production along the way. Their stories gave me the courage to start rapping as well. I began by imitating and covering famous songs. I also experimented freestyling with my friends, saying whatever came to mind and felt right within the rhythm. I also learned DJing and beatboxing on the way, before I eventually began making individual and collaborative rap demos with my best friends.

These songs are my diaries. They record my authentic life experiences, honest feelings, and opinions. I believe there's something especially personal about making and sharing music, where the words I write on paper become infused with my voice and raw emotions. It's like opening my soul for others to see.

Describe the most significant challenge you have faced and the steps you have taken to overcome this challenge. How has this challenge affected your academic achievement?

In the summer of 2021, I attended Pioneer Academics, an online research program where I spent three months learning about Fourier Analysis and writing a research paper on it. The first five weeks were spent building up the mathematical foundations for Fourier Analysis. The professor would host a two-hour group session every Saturday to lecture us on the topic. Then, throughout the week, we would work on his assigned problems.

Those first few weeks of studying were extremely challenging for me, as Fourier Analysis is an advanced concept that requires much prerequisite knowledge such as differential equations, linear algebra, and real analysis, all of which I didn't know at the time. Furthermore, two hours of class time is nowhere near enough for the professor to fully explain these complicated concepts to us. Luckily, self-study is one of my specialties, and I managed to work through all the textbooks and YouTube lectures to gain more insight into these concepts. In those first weeks, I spent more than five hours each day self-studying and completing homework assignments.

Afterward, we initiated individual sessions, once a week, to discuss my research paper. Inspired by the contemporary circumstances and news on the coronavirus variants, I decided to analyze the periodicity of COVID-19. Fourier Transform is great at analyzing periodicities, and the research result is relevant and meaningful, potentially aiding the government in planning public health interventions. My professor also gave me many useful suggestions on the general direction of my investigation and ideas for the representation of data. Eventually, I produced a 20+ page research report that commented on the weekly, annual, and biannual periodicities found in COVID-19, and received an A grade.

Through this program, I not only experienced the research process and picked up advanced mathematics, physics, and computer science knowledge, but I also learned to appreciate the importance and power of STEM in initiating insight into policy making. As someone interested in interdisciplinary studies and incorporating STEM with humanities, I am thankful that this experience provided me the motivation to continue such work in the future.

What have you done to make your school or your community a better place?

I love to be outdoors, running, cycling, playing soccer, and keeping my fitness levels up. It brings me confidence, happiness, and so much more to list. Knowing how much fitness and health matter to my life, I hoped to make them more accessible to others. So, last summer, I became an intern at Yunyou, a tech company in China, where I could intertwine my love for sports and my service goals with my STEM background to help create wearable devices that support people's fitness goals. The company uses the world's leading technologies-such as conductive fiber material, artificial intelligence, and big data-to build a large health management platform that utilizes intelligent clothing to monitor human vital signs.

At Yunyou, I worked in the algorithm department to develop signal processing and artificial intelligence waveform recognition for an electrocardiogram (ECG) tool that would be incorporated into a smart-clothing design. As part of my work, I began learning about the structure of the human heart and the technical aspects of an ECG. After weeks of intense study in biology and sports science, I eventually joined the signal processing subdepartment to work on reducing noise interference in ECG signals. Finally, I created a long short-term memory (LSTM) neural network that identifies waveforms in ECG signals. The labeled results were then used by my colleagues to help diagnose heart diseases or sub-optimal health levels.

I learned much about biology, engineering, and their intersection with public health. But what really appealed to me about this work was that our products can really help customers improve their health awareness, better manage their body data, and become healthier. Through subsidies and fundraising, our devices would support underserved communities that lack medical resources. Now, elders in rural villages have access to the ECG we developed and can live their lives with improved health awareness. By detecting medical-grade ECG signals and warning users of potential heart risks, our device significantly lowers the risk of cardiovascular diseases in these communities, and I am proud to have been part of this work.

Additional comments

1. Track & Field Club:
I have always been incredibly passionate about track and field. My 100m time was 11.7 seconds, which is equal to a Chinese national second-level athlete. I love track and field because I love the extreme self-challenge, involving pure hard work, pure commitment, and rebuilding yourself through hardship that leads to the

courage to face adversity. However, there is no varsity track and field at my school, so I am currently communicating with the school developing one with a coach from Shanghai Professional Sports School and more funding. Hopefully I can build a track and field school team next semester.

2. How I Started Teaching at Ganluo Middle School:
I heard of this school because its headmaster was my father's primary school classmate. 95% of the students in that school were of the Yi Minority, and less than 40% went to high school. Born in a poor village close to Ganluo, I felt obliged to do something to support this community since I knew how hard life was there. I thought sports could be the way, since they may not end up going to college or speaking English, but the enthusiasm for life that sports brings might ease some of the hardships they experience and give them another positive in their daily life. This is why I started volunteer teaching at the school and helping them build basketball courts.

3. Abstract of the Pioneer Research Paper:
The purpose of this study was to perform frequency domain analysis on COVID-19 data, and potentially aid the government in planning public health interventions. COVID-19 time series were transformed into the frequency domain and plotted on periodograms for data visualization. In this study, weekly, annual, and biannual periodicities were found, and this paper provides interpretations and explanations of each periodicity. This paper also presents a detailed explanation and derivation of Discrete Fourier Transform Formula and Fast Fourier Transform Algorithm.

4. Abstract of the Research on the Relationship between Regional Poverty and Sports Development Level:
Sports have long proven to be a route out of poverty, providing individuals with a path to college and generating revenue for states from competition and advertisement. However, the hardships that athletes must go through in order to truly make it out of poverty are immense, especially in physical-training-oriented sports like track and field, weightlifting, and gymnastics. Meanwhile, some theories suggest that youth from low income households would prevail in these sports, as kids from higher income households might not wish to undertake the physical challenge when faced with better life options. Inspired by this claim, this study looks to reveal the statistical relationship between regional poverty and sports development level, and potentially aid the government in planning sports poverty alleviation policies. The Stata data analysis results reveal a statistically positive correlation between poverty and athletic development. This paper also provides a detailed explanation of the analytical results with underdog theory.